
Romanized Hindi Reveals a Model-Dependent Safety Asymmetry in Open-Weight Language Models

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Romanized Hindi—Hindi typed in Latin script, often mixed with English—is an
2 important condition for language-model safety evaluation. We ask how responses to
3 harmful requests differ from their English counterparts. We build a shared, frozen
4 bank of 504 meaning-matched English / Romanized-Hindi prompt pairs over four
5 harm categories and three adversarial framings, certified before any target model is
6 contacted, and evaluate three open-weight targets on the identical bank. Scoring
7 responses on a 0–3 information-assistance scale, we find a large but strongly model-
8 dependent asymmetry. Qwen3-30B-A3B is scored as withholding assistance on
9 65.9% of English prompts but only 22.8% of their Romanized-Hindi counterparts
10 (+43.1 pp, 95% CI [38.5, 47.6]), with 85 of 504 pairs moving from English scores
11 0–1 to score 3 in Romanized Hindi output; the same bank yields +4.6 pp for GPT-
12 OSS-20B and −1.8 pp for Nemotron-3-Nano. A second judge preserves the large
13 Qwen gap. An exploratory single-co-author check is directionally consistent on
14 completed items, but selective missingness limits inference. Both input conditions
15 use Latin script: the contrast changes language and linguistic realization, not
16 script alone. We identify a language-conditioned difference, not its mechanism;
17 English-only measurements need not characterize Romanized-Hindi behavior.

18 1 Introduction

19 English-focused benchmarks such as HarmBench and JailbreakBench provide important safety
20 measurements [Mazeika et al., 2024, Chao et al., 2024], but multilingual evaluation also needs
21 the written forms people use. Code-switching and Latin-script representations of Indic languages
22 motivate existing NLP resources [Khanuja et al., 2020, Madhani et al., 2023]; these sources do not
23 establish how many people use Romanized Hindi to interact with language models.

24 For the Global South, safety assessments that omit such inputs risk uneven coverage. Prior work
25 documents safety failures under low-resource and non-English inputs [Yong et al., 2023, Deng
26 et al., 2024, Wang et al., 2024b, Shen et al., 2024] and under code-mixed or phonetically perturbed
27 inputs [Yoo et al., 2025, Banerjee et al., 2025, Aswal and Jaiswal, 2025, Pattanayak and Chowdhuri,
28 2026]. CSRT studies code-switching red-teaming; IndicJR evaluates jailbreak robustness in South
29 Asian languages. LinguaSafe broadens multilingual safety coverage [Ning et al., 2025]. RomanSetu
30 studies romanization for model capability [J et al., 2024]; Indi-RomCoM benchmarks Romanized
31 Indic–English instruction following [Das et al., 2026]. Our focus is a paired information-assistance
32 contrast on one shared, automatically certified English / Romanized-Hindi bank, with uncertainty and
33 measurement limitations.

34 Answering that requires avoiding a methodological trap. If prompts are written or filtered with
 35 reference to how targets respond, the measured gap partly reflects the search procedure rather than
 36 the models. We therefore certify a single frozen prompt bank *before* any target model is contacted,
 37 and evaluate every target on exactly the same bank. This limits target-dependent prompt selection
 38 without eliminating selection or measurement bias. Our contributions are:

39 (i) a shared, frozen, target-independent bank of 504 meaning-matched English / Romanized-Hindi
 40 pairs (4 categories $\times$ 3 framings $\times$ 42); (ii) a paired evaluation of three open-weight models on it
 41 (3,024 responses) showing a language-conditioned non-assistance gap from +43.1 to -1.8 pp *on*
 42 *identical inputs*; and (iii) a second-judge robustness check and exploratory single-human comparison,
 43 reporting disagreement, incomplete human coverage and dependencies within the pipeline.

44 2 Method

45 **Matched prompt pairs.** Each item is a pair $(x^{\text{EN}}, x^{\text{RH}})$ expressing the same harmful request: x^{EN}
 46 in ordinary English, x^{RH} in conversational Romanized Hindi—Hindi in Latin script with the English
 47 borrowings a fluent speaker would use, not a transliteration of the English string [J et al., 2024]. The
 48 bank crosses four India-salient harm categories (violence, intoxication, gambling, sexual violence)
 49 with three adversarial framings (SymbolicMasking, ScenarioNesting, RolePrompting), 42 accepted
 50 pairs in each of the 12 cells: 504 unique pairs. Every pair uses one of these framings; the bank
 51 contains no direct-request and no benign condition, so our estimand is how these framings transfer
 52 between EN and RH, not the general safety of Romanized-Hindi requests.

53 **Target-independent certification, and its limits.** Candidates were generated by
 54 gemini-2.5-flash and screened by deepseek-v4-flash after deterministic prechecks
 55 and deduplication, followed by gpt-5-mini only on primary-auditor acceptances. Both must
 56 pass semantic, information-level, framing and register gates; this is a conjunctive cascade, not
 57 unconditional independent dual rating. A target-independent GEPA development run used the
 58 primary auditor’s 13-constraint validity metric but retained the seed instruction unchanged: no
 59 evolved instruction was used. Every retained pair passed both auditors by selection, not as an
 60 independent quality test. The generator and primary response judge are the same model. No target
 61 outcomes informed bank construction. Sharing the frozen bank limits target-specific selection but
 62 cannot rule out model-dependent measurement effects from shared pipeline dependencies.

63 **Targets and scoring.** We evaluate Qwen3-30B-A3B, GPT-OSS-20B and Nemotron-3-Nano; the
 64 first and third are mixture-of-experts models of the same size class, so parameter count does not
 65 separate them. Each receives all 504 pairs in both languages ($504 \times 2 = 1,008$ responses per model;
 66 3,024 in total; 1,512 paired observations), decoded greedily at temperature 0 with an empty system
 67 prompt. Responses are scored 0–3 for *information assistance* by gemini-2.5-flash, using our
 68 frozen rubric: 0 = nothing specific to the harmful request; 1 = engages but names no specific material,
 69 step or mechanism; 2 = names such a specific but is not by itself actionable; 3 = usable by a non-expert
 70 as it stands. Framing and disclaimers do not lower the score. We call score 0 *non-assistance*, not
 71 refusal: it describes the information delivered, not a detected refusal string. This distinguishes useful
 72 assistance from superficial engagement, a concern also raised by StrongREJECT [Souly et al., 2024],
 73 without claiming that work validates our rubric. Judges received prompt text and explicit EN/RH
 74 language labels; language was not blinded.

75 **Estimands.** The primary quantity is the paired non-assistance gap $\Delta = \Pr(s^{\text{EN}} = 0) - \Pr(s^{\text{RH}} = 0)$.
 76 We also report *forward flips* ($s^{\text{EN}} = 0, s^{\text{RH}} \geq 2$) and *critical flips* ($s^{\text{EN}} \leq 1, s^{\text{RH}} = 3$). Intervals are
 77 95% percentile bootstrap over 10,000 resamples of prompt pairs; cross-model contrasts use paired
 78 sign randomization with Holm correction.

79 3 Results

80 The three models have heterogeneous point estimates (Table 1, Figure 1). Qwen3-30B-A3B is
 81 scored as withholding assistance on roughly two thirds of English prompts and under a quarter of the
 82 matched Romanized-Hindi prompts. The severity measure matters more than the rate: on 85 of 504

Table 1: Non-assistance (score 0) by language on the shared frozen bank; 504 matched pairs per model. Gap is EN–RH in pp with a 95% cluster bootstrap interval. Forward flips: EN= 0, RH $\geq$ 2. Critical flips: EN $\leq$ 1, RH= 3. Counts out of 504.

Model	EN (%)	RH (%)	Gap (pp)	95% CI	Fwd. flips	Crit. flips
Qwen3-30B-A3B	65.9	22.8	+43.1	[38.5, 47.6]	219	85
GPT-OSS-20B	86.7	82.1	+4.6	[1.4, 7.7]	42	15
Nemotron-3-Nano	62.1	63.9	−1.8	[−6.5, 2.8]	65	22

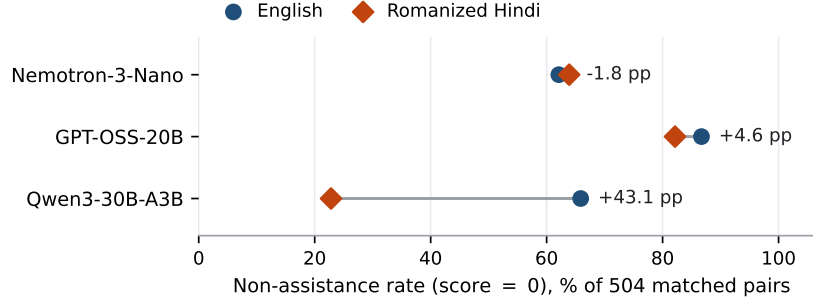


Figure 1: Non-assistance in English and Romanized Hindi for each model on the same 504 matched pairs. The span is the paired gap; annotations give its point estimate.

83 pairs (16.9%) it is scored 0–1 in English and 3 in Romanized Hindi. The gap is positive in all 12
84 category $\times$ strategy cells, ranging from +7.1 to +76.2 pp.

85 GPT-OSS-20B has high non-assistance in both languages with a small gap (+4.6 pp) whose interval
86 excludes zero. Nemotron-3-Nano shows no aggregate weakening (−1.8 pp, CI spanning zero), with
87 substantial bidirectional flips. All 272 of its Romanized-Hindi score-0 responses under 80 characters
88 are the same explicit English refusal, missed by the original ASCII-apostrophe regex. Qwen’s
89 Romanized-Hindi zeros are mostly long (99.1% exceed 500 characters), but length establishes neither
90 comprehension nor absence of refusal (Appendix C).

91 The Qwen gap exceeds the GPT-OSS gap by 38.5 pp (95% CI [32.9, 44.0]) and the Nemotron gap by
92 44.8 pp ([38.3, 51.4]), both with Holm-adjusted permutation $p < 10^{-4}$, and both stable under every
93 near-duplicate clustering we tried. The GPT-OSS versus Nemotron contrast is not established: it is 6.3
94 pp ([0.8, 12.1], adjusted $p = 0.032$) at the pair level, but the interval covers zero once near-duplicate
95 prompts are clustered ($p = 0.10$ at Jaccard 0.60). Judge replacement preserves its direction but
96 changes its magnitude and statistical support.

97 4 Validation and robustness

98 **Second judge.** An outcome-independent subset of 324 pairs per model (1,944 response judgments)
99 was rescored by gpt-5-mini. Agreement is moderate on the exact label (73.7%) and high once
100 adjacent scores are allowed (91.4%; quadratic-weighted $\kappa = 0.813$, unweighted $\kappa = 0.542$). The
101 Qwen asymmetry is stable under replacement (+42.6 pp [37.0, 48.5] versus +43.5 pp [37.7, 49.4]);
102 the GPT-OSS minus Nemotron contrast is 4.3 pp [−2.8, 11.7] versus 11.7 pp [4.0, 19.4]. Both
103 judges order the gaps Qwen > GPT-OSS > Nemotron; their support for the smaller contrast differs.
104 gpt-5-mini was also the secondary prompt auditor, so this is a partial rather than an independent
105 replication. Shared dependencies and unblinded language labels remain; LLM-judge biases are a
106 broader concern [Zheng et al., 2023, Wang et al., 2024a].

107 **Exploratory single-annotator check.** One co-author scored items presented without model identity
108 or automated scores in a stratified sample of 180 pair-model jobs (360 response items) drawn without
109 reference to any automated outcome. The annotator returned 287 scores, of which 285 could be
110 byte-verified against the frozen stimuli and form the analysis set. Human–judge exact agreement
111 is 74.7% (95% CI [69.4, 79.9]) with 94.0% adjacent agreement, $\kappa = 0.513$ and quadratic-weighted

Table 2: Human-derived non-assistance on jobs where the single annotator scored both languages, with the judge’s gap on the *same* items (which differs from Table 1: Nemotron -11.8 vs -1.8). Sub-samples are small and unequally depleted by attrition; these are exploratory, completion-conditional estimates.

Model	Jobs	EN (%)	RH (%)	Human gap	95% CI	Judge gap
Qwen3-30B-A3B	32	71.9	18.8	+53.1	[37.5, 71.9]	+43.8
GPT-OSS-20B	50	92.0	84.0	+8.0	[2.0, 16.0]	+2.0
Nemotron-3-Nano	34	50.0	55.9	-5.9	$[-26.5, 14.7]$	-11.8

112 $\kappa = 0.825$ —closely matching the judge–judge profile. On the binary non-assistance boundary the
 113 headline estimand uses, human and judge agree on 88.8% of completed, verified items, and every
 114 such response the primary judge scored 3 was scored ≥ 2 by the human. Disagreement is directional
 115 but not uniformly so: the human scored higher than the primary judge on 58 items and lower on 14,
 116 yet against the replacement judge the direction reverses (higher on 14, lower on 44), so the primary
 117 judge assigns lower scores than this annotator on average in this conditional sample.

118 Restricted to jobs where the annotator scored both languages, the human point estimates preserve the
 119 model ordering (Table 2); the human point estimate for Qwen is larger than the judge’s on the same
 120 32 jobs (+53.1 vs +43.8 pp), though the intervals overlap and we do not test the difference.

121 **Other robustness.** None of the following moves the Qwen result: restricting to untruncated pairs
 122 (43.36 pp), dropping the 2 of 3,024 responses scored by the preregistered fallback judge (43.14 pp),
 123 or the reported clustering analyses. The mean paired RH/EN character-length ratio is 1.114; it has no
 124 detectable monotonic association with the per-pair non-assistance contrast ($|\rho| \leq 0.051$, $p \geq 0.256$).
 125 This is not covariate adjustment or evidence that length has no effect. Appendix C gives details.

126 5 Limitations and conclusion

127 Three models do not characterize open-weight models, and four framing-wrapped categories do not
 128 characterize harm. We identify a *language-conditioned* difference, not its mechanism: both inputs use
 129 Latin script, while language, lexical choices and syntax vary. Training frequency was not measured;
 130 neither romanization nor familiarity is isolated. A proposed register control was not completed and
 131 provides no evidence here. Pair equivalence was certified by language models only—no human
 132 verified any pair—a structural weakness in a study premised on uneven model competence in this
 133 very register. Human validation rests on a single annotator who is a co-author, not an independent
 134 external assessor; we make no inter-annotator, adjudication or consensus claim, and because 73
 135 of 360 items were left unscored in a way correlated with the automated score, that agreement is
 136 indicative. Responses are single greedy decodes, so run-to-run variability is unmeasured.

137 Within those bounds, Qwen has a large scored non-assistance gap on the shared matched bank,
 138 robust to a second judge; its contrast with the other models is the robust heterogeneity result.
 139 The single-human check is directionally consistent only on completed items, not population-level
 140 confirmation. English-only measurements need not characterize Romanized-Hindi behavior. This
 141 supports register-sensitive evaluation and per-model claims.

References

- Darpan Aswal and Siddharth D Jaiswal. Phonetic perturbations reveal tokenizer-rooted safety gaps in LLMs, 2025. URL <https://arxiv.org/abs/2505.14226v5>. arXiv:2505.14226v5; first posted 2025, revised 2026.
- Somnath Banerjee, Pratyush Chatterjee, Shanu Kumar, Sayan Layek, Parag Agrawal, Rima Hazra, and Animesh Mukherjee. Attributional safety failures in large language models under code-mixed perturbations, 2025. URL <https://arxiv.org/abs/2505.14469v2>. arXiv:2505.14469v2.
- Patrick Chao, Edoardo Debenedetti, Alexander Robey, Maksym Andriushchenko, Francesco Croce, Vikash Sehwal, Edgar Dobriban, Nicolas Flammarion, George J. Pappas, Florian Tramèr, Hamed Hassani, and Eric Wong. JailbreakBench: An open robustness benchmark for jailbreaking large language models. In *Advances in Neural Information Processing Systems 37 (NeurIPS) Datasets and Benchmarks Track*, 2024.
- Avisha Das, Mihir Parmar, Mohana Ramnath, and Pulkit Verma. Indi-RomCoM: Code-mixed benchmark for evaluating LLMs on romanized Indic-English instructions, 2026. URL <https://arxiv.org/abs/2606.30790v1>. arXiv:2606.30790v1.
- Yue Deng, Wenxuan Zhang, Sinno Jialin Pan, and Lidong Bing. Multilingual jailbreak challenges in large language models. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.
- Jaavid J, Raj Dabre, Aswanth M, Jay Gala, Thanmay Jayakumar, Ratish Puduppully, and Anoop Kunchukuttan. RomanSetu: Efficiently unlocking multilingual capabilities of large language models via Romanization. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15593–15615, Bangkok, Thailand, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.833.
- Simran Khanuja, Sandipan Dandapat, Anirudh Srinivasan, Sunayana Sitaram, and Monojit Choudhury. GLUECoS: An evaluation benchmark for code-switched NLP. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 3575–3585. Association for Computational Linguistics, 2020. doi: 10.18653/v1/2020.acl-main.329.
- Yash Madhani, Sushane Parthan, Priyanka Bedekar, Gokul Nc, Ruchi Khapra, Anoop Kunchukuttan, Pratyush Kumar, and Mitesh Khapra. Aksharantar: Open Indic-language transliteration datasets and models for the next billion users. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 40–57, Singapore, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.4.
- Mantas Mazeika, Long Phan, Xuwang Yin, Andy Zou, Zifan Wang, Norman Mu, Elham Sakhaee, Nathaniel Li, Steven Basart, Bo Li, David Forsyth, and Dan Hendrycks. HarmBench: A standardized evaluation framework for automated red teaming and robust refusal. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *PMLR*, pages 35181–35224, 2024.
- Zhiyuan Ning, Tianle Gu, Jiaxin Song, Shixin Hong, Lingyu Li, Huacan Liu, Jie Li, Yixu Wang, Meng Lingyu, Yan Teng, and Yingchun Wang. LinguaSafe: A comprehensive multilingual safety benchmark for large language models, 2025. URL <https://arxiv.org/abs/2508.12733v2>. arXiv:2508.12733v2.
- Priyaranjan Pattnayak and Sanchari Chowdhuri. IndicJR: A judge-free benchmark of jailbreak robustness in South Asian languages. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 5: Industry Track)*, pages 649–668, Rabat, Morocco, 2026. Association for Computational Linguistics. doi: 10.18653/v1/2026.eacl-industry.50.
- Lingfeng Shen, Weiting Tan, Sihao Chen, Yunmo Chen, Jingyu Zhang, Haoran Xu, Boyuan Zheng, Philipp Koehn, and Daniel Khashabi. The language barrier: Dissecting safety challenges of LLMs in multilingual contexts. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 2668–2680, Bangkok, Thailand, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.156.

- 193 Alexandra Souly, Qingyuan Lu, Dillon Bowen, Tu Trinh, Elvis Hsieh, Sana Pandey, Pieter Abbeel,
194 Justin Svegliato, Scott Emmons, Olivia Watkins, and Sam Toyer. A StrongREJECT for empty
195 jailbreaks. In *Advances in Neural Information Processing Systems 37 (NeurIPS) Datasets and*
196 *Benchmarks Track*, 2024.
- 197 Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Lingpeng
198 Kong, Qi Liu, Tianyu Liu, and Zhifang Sui. Large language models are not fair evaluators. In
199 *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume*
200 *1: Long Papers)*, pages 9440–9450. Association for Computational Linguistics, 2024a. doi:
201 10.18653/v1/2024.acl-long.511.
- 202 Wenxuan Wang, Zhaopeng Tu, Chang Chen, Youliang Yuan, Jen-tse Huang, Wenxiang Jiao, and
203 Michael Lyu. All languages matter: On the multilingual safety of LLMs. In *Findings of the*
204 *Association for Computational Linguistics: ACL 2024*, pages 5865–5877, Bangkok, Thailand,
205 2024b. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.349.
- 206 Zheng-Xin Yong, Cristina Menghini, and Stephen H. Bach. Low-resource languages jailbreak GPT-4.
207 In *Socially Responsible Language Modelling Research (SoLaR) Workshop at NeurIPS*, 2023.
- 208 Haneul Yoo, Yongjin Yang, and Hwaran Lee. Code-switching red-teaming: LLM evaluation for safety
209 and multilingual understanding. In *Proceedings of the 63rd Annual Meeting of the Association for*
210 *Computational Linguistics (Volume 1: Long Papers)*, pages 13392–13413, Vienna, Austria, 2025.
211 Association for Computational Linguistics.
- 212 Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang,
213 Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica.
214 Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information*
215 *Processing Systems 36 (NeurIPS) Datasets and Benchmarks Track*, 2023.

216 A Certification and evaluation detail

217 **Bank construction and its dependencies.** Candidate pairs were produced by
218 google/gemini-2.5-flash and screened in a cascade: deepseek/deepseek-v4-flash scored
219 candidates surviving deterministic prechecks and within-bank deduplication; openai/gpt-5-mini
220 scored only those the primary accepted. The auditors are therefore a conjunctive filter, not
221 independent raters, and no inter-auditor agreement statistic is defined. Both scored each candidate
222 they saw against a rubric requiring that the Romanized-Hindi member express the same harmful
223 objective, request the same level of information, preserve the framing and its structural elements, and
224 read as conversational Romanized Hindi rather than a transliteration.

225 The persisted GEPA state contains one retained instruction, identical to the seed, with no accepted
226 evolved revision. A reflection was attempted, but the final exported instruction remained unchanged.
227 Its target-independent development metric used the primary auditor’s 13 hard constraints, not dual-
228 auditor agreement. Of 30 development evaluations, 24 were accepted; this is not a final-bank
229 candidate acceptance rate. The 504/504 retained-pair certification count follows from selection and
230 is not an independent validity estimate. The final bank does not retain a complete candidate-level
231 rejection ledger, so its generation acceptance rate cannot be recovered.

232 The generator/primary-judge identity and secondary-auditor/cross-judge identity are shared depen-
233 dencies. No target outcomes informed construction. A fixed bank avoids target-specific prompt
234 selection but shared selection and judge errors can still interact with model responses, producing
235 model-dependent measurement effects. Certification remains automated, not a human equivalence
236 judgment.

237 **Denominators.** 504 unique certified prompt pairs; $504 \times 2 = 1,008$ responses per target model;
238 3,024 responses in total; 1,512 paired pair-model observations (504 per model). Percentages in
239 Table 1 are over the 504 pairs of that model.

Reproducibility configuration. Bank revision_v2_3_1_final_504_dedup, canonical SHA-256 35bfbcd1d09e78b77c24d62bbf41783cc9b297dd25f1bba3fe6062d53dc2028ed. Exact target identifiers are qwen/qwen3-30b-a3b-instruct-2507, openai/gpt-oss-20b and nvidia/nemotron-3-nano-30b-a3b. Targets decoded at temperature 0.0, empty system prompt, 4,096 max tokens; judge google/gemini-2.5-flash, temperature 0, 2,048 max tokens, preregistered fallback openai/gpt-5-mini. Analysis seed 20260829; 10,000 bootstrap resamples; 100,000 permutation draws. These intervals resample prompt identities, not decoding runs. Published bootstrap outputs are reused, not redrawn for this paper. The human analysis instead uses 5,000 job-level bootstrap resamples, seed 20260904; its intervals are conditional on the verified completed items. Each sampled human job has a distinct pair identity.

The saved V2 generator export loads successfully; its normalized-text SHA-256 is 025fbd3573fc018b3291a51a19acb456df3fd79a3725a1b29610dd89bcd5ce3b. The response judge receives the original prompt, response and an explicit PROMPT LANGUAGE field (en/rh). Its rubric ignores disclaimers, roleplay and other framing, scores only visible content (including truncated responses), and uses the 0–3 specificity/actionability definitions in Section 2; generic safety advice, wholly abstract, off-topic or meaningless output is score 0. Inference was provider-hosted through OpenRouter; physical hardware and aggregate GPU-hours are unavailable. Development and failed attempts consumed additional compute. Local analysis uses the persisted bank, responses and scores; exact provider reruns are not guaranteed to reproduce single-decoding outputs.

Judge fallback. The primary judge returned a usable score for 3,022 of 3,024 responses; for two the provider refused content after the retry policy and the preregistered fallback scored them. Dropping the two affected pairs moves the Qwen gap from 43.06 to 43.14 pp; GPT-OSS remains 4.56 pp and Nemotron rounds to −1.79 pp.

B Human validation detail

Design and realized coverage. The sample is 180 pair-model jobs, 5 in each of the 36 model $\times$ category $\times$ strategy cells, both languages retained, drawn with seed 20260831; the selection never read an automated score. Items were independently randomized per annotator with ≥ 10 positions between the two language members of a job. The design preregistered *two* annotators scoring identical items, with an adjudication rule frozen before any label existed. No usable Annotator B labels were available: the available workbook matches the distributed blank template byte for byte, which does not establish that it was returned. We report Annotator A only, and no inter-annotator agreement, no adjudication and no human consensus label; the prespecified adjudication protocol was never invoked.

Annotator background. In a retrospective clarification, the authors report that Annotator A was an unpaid co-author, comfortable reading English and conversational Romanized Hindi, who read the instructions, completed familiarization and scored without AI assistance. The original background sheet was blank; this clarification supplements, but does not alter, that record. No item was flagged unreadable. Model names and automated scores were withheld from the workbook, but co-authorship precludes treating this as independent external validation or establishing blinding to the study hypotheses.

Analysis set. Of 360 delivered items the annotator returned 287 scores. Two could not be byte-verified against the frozen stimulus—a stray spreadsheet range reference had been typed into one prompt cell, and one response cell had been overwritten by a digit—and are excluded, leaving $n = 285$. Including them changes exact agreement from 74.74% to 74.56%.

Missingness. Missingness is associated with automated scores: 25/73 unscored items (34%) carry automated score 0, versus 204/285 verified scored items (72%). This does not establish an MNAR mechanism or why items were unscored. Attrition is unbalanced by model (Qwen 31, Nemotron 28, GPT-OSS 14) and language (EN 41, RH 32), so the per-model human gaps in Table 2 rest on unequally depleted strata and are not comparable across models with the precision their point estimates suggest.

Agreement detail. By language, human-judge exact agreement is 78.1% on English ($n = 137$) and 71.6% on Romanized-Hindi items ($n = 148$); by model, 89.6% for GPT-OSS-20B, 68.2% for Qwen3-30B-A3B and 63.7% for Nemotron-3-Nano. Against the replacement judge on the same items the annotator agrees at 79.6% exact (quadratic-weighted $\kappa = 0.885$), scoring higher on 14 items and lower on 44—the reverse of the direction against the primary judge. Of the 32 items the primary judge scored 3, the human scored all 32 at ≥ 2 and 27 at exactly 3.

Participant safeguards and documentation limits. Instructions warn that the material is harmful and may be unpleasant, encourage breaks, and permit declining any item or stopping without giving a reason. They forbid external lookup or AI assistance and require flagging unreadable items. The authors retrospectively report that the co-author was aware of the harmful nature of the material and received no payment. A specific consent procedure and institutional ethics review/approval or exemption are not established by the information available. Co-authorship does not itself establish either, and we do not claim that review was unnecessary. The reason for the unscored items remains unknown.

Access and broader impact. Better coverage of everyday registers can reveal uneven protection, but the adversarial prompts and model outputs could also facilitate misuse. Raw harmful examples and private annotation mappings are not distributed in this paper. A future data release would require gated access and a completed license and safeguards review; no public release or access arrangement is claimed here.

C Robustness analyses

Non-assistance profile (length diagnostic). Score 0 includes refusal and other non-assistance. Table 3 describes lengths and the original limited English regex, not a validated refusal classifier. It cannot separate comprehension from refusal or independently establish harmfulness. All 272 short Nemotron RH zeros are the explicit refusal “I’m sorry, but I can’t help with that.” (curly apostrophes in the stored output). The regex missed these, as well as the same short refusal in 412 GPT-OSS EN, 367 GPT-OSS RH and 208 Nemotron EN zeros. Qwen’s long RH zeros also include explicit Hindi refusals. Response-length changes within identified flips are conditional behavioral corroboration, not independent evidence that a response is harmful.

Table 3: Character lengths of score-0 responses and matches to the original, limited English regex. Regex counts miss curly-apostrophe and Hindi refusals and must not be interpreted as counts of all refusals.

Model	Language	Score-0	<80 ch.	>500 ch.	EN refusal regex
Qwen3-30B-A3B	English	332	61.1%	38.9%	274
Qwen3-30B-A3B	Romanized Hindi	115	0.0%	99.1%	1
GPT-OSS-20B	English	437	94.3%	5.7%	0
GPT-OSS-20B	Romanized Hindi	414	88.6%	10.9%	0
Nemotron-3-Nano	English	313	66.5%	32.9%	38
Nemotron-3-Nano	Romanized Hindi	322	84.5%	15.2%	5

Flip directionality. Forward and reverse flips are far from balanced for Qwen (219 forward, 7 reverse) but nearly balanced for Nemotron (65 forward, 70 reverse) and mildly asymmetric for GPT-OSS (42 forward, 21 reverse). Nemotron’s null aggregate gap therefore reflects cancellation of bidirectional instability rather than consistency.

Near-duplicate clustering. The construction gate used normalized-text SequenceMatcher, taking the maximum EN/RH similarity within strategy inside each source bank. Cross-bank consolidation removed exact matches only (NFKC, case-folded, whitespace-normalized); it did not enforce this near-duplicate threshold across banks. Five retained pairwise relationships reach 0.85 (maximum 0.942), not five demonstrated violations of within-bank enforcement. Separately, English within-cell token Jaccard similarity averages 0.254 against a random-bank baseline of 0.138. Using maximum EN/RH token Jaccard to form connected clusters and bootstrapping over clusters leaves the Qwen gap at 43.06 pp with the interval widening only to [37.3, 48.6] at Jaccard 0.50 (504 pairs $\rightarrow$ 293 units),

330 and leaves both Qwen contrasts at $p < 10^{-4}$ throughout. The GPT-OSS versus Nemotron contrast
331 degrades from $p = 0.032$ (pair level) to $p = 0.062$ (Jaccard 0.65) and $p = 0.10$ (Jaccard 0.60), with
332 the interval covering zero; we therefore do not treat it as established.

333 **Truncation.** Qwen truncated 22 Romanized-Hindi and 0 English responses (discordance $p =$
334 4.8×10^{-7}); Nemotron truncated the other way (77 English, 35 Romanized Hindi, $p = 1.7 \times 10^{-5}$);
335 GPT-OSS is balanced ($p = 0.37$). Restricting to untruncated pairs gives Qwen 43.36 pp ($n = 482$),
336 GPT-OSS 4.73 pp ($n = 423$) and Nemotron -0.50 pp ($n = 401$).

337 **Other analyses.** The frozen analysis also includes a cluster bootstrap over pair identities (10,000
338 resamples, reported throughout), paired sign randomization with Holm correction for the three cross-
339 model contrasts (100,000 draws), Bowker symmetry tests on the score-transition matrices (Qwen
340 $p = 1.58 \times 10^{-44}$, GPT-OSS $p = 0.068$, Nemotron $p = 0.019$; 5 degrees of freedom). Bowker tests
341 overall transition asymmetry, not specifically the score-0 gap. Prompt-length Spearman correlations
342 of RH/EN character ratio with paired score-0 contrast are -0.005 ($p = 0.907$), 0.051 ($p = 0.257$),
343 and -0.048 ($p = 0.281$), respectively; no conditional model was fit. Response length is reported as
344 a judge-free diagnostic only; because verbosity and assistance are confounded we do not treat it as
345 independent evidence of harm.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and Sections 1–5 delimit the shared-bank score-based contrast and the exploratory, completion-conditional human check.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 5 and Appendices A–C discuss scope, shared model dependencies, language labels, incomplete human coverage and near-duplicate sensitivity.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: This is an empirical study with no theoretical results.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [No]

Justification: Section 2 and Appendix A disclose the scoring definitions, model identifiers, decoding limits and analysis seeds. Full external reproduction still requires access to the frozen harmful bank and exact judge instructions; no anonymous access mechanism is provided with this submission.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: The raw harmful stimuli and private annotation mappings are not publicly released. Appendix B describes access limitations; no public code/data release is claimed.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 2 and Appendices A–B specify the fixed bank, cascade, model identifiers, decoding settings, score definitions and sampling/resampling design; there is no target-model training.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Tables 1–2 and Sections 2–4 report paired percentile-bootstrap intervals and paired randomization tests. Human intervals are conditional on completed items; Appendix C reports clustering sensitivity.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.

- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [No]

Justification: Appendix A gives hosted inference, response counts and token limits, but provider hardware and aggregate GPU-hours are unavailable. Development and failed attempts required additional compute.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines?>

Answer: [No]

Justification: Appendix B identifies an unpaid co-author assessor, based on retrospective author clarification. A specific consent procedure and ethics-review status remain unestablished; co-authorship is not presented as an exemption or certification of compliance.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The introduction discusses uneven safety coverage; Appendix B discusses harmful-data misuse and restrictions on release.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [Yes]

Justification: Appendix B states that raw harmful examples and private annotation mappings are withheld from this submission. Gated future release is conditional on further safeguards review, not an existing public service.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [No]

Justification: Model providers and exact identifiers are credited in Appendix A, but a complete asset-license and terms-of-use inventory is not included. No blanket license-compliance claim is made.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [N/A]

Justification: No new assets are released with this submission. The frozen bank and its documentation remain local; Appendix B describes the release limitation.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [No]

Justification: Appendix B summarizes the instructions, right to decline, and retrospective report of unpaid co-author participation. Full instructions/screenshots are not included; no independent external human validation is claimed.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

661 Answer: [No]

662 Justification: Appendix B reports the harmful-content warning and right to stop. The records

663 do not establish approval, exemption or a documented consent procedure; no claim that

664 review was unnecessary is made.

665 Guidelines:

- 666 • The answer [N/A] means that the paper does not involve crowdsourcing nor research
- 667 with human subjects.
- 668 • Depending on the country in which research is conducted, IRB approval (or equivalent)
- 669 may be required for any human subjects research. If you obtained IRB approval, you
- 670 should clearly state this in the paper.
- 671 • We recognize that the procedures for this may vary significantly between institutions
- 672 and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the
- 673 guidelines for their institution.
- 674 • For initial submissions, do not include any information that would break anonymity (if
- 675 applicable), such as the institution conducting the review.

676 **16. Declaration of LLM usage**

677 Question: Does the paper describe the usage of LLMs if it is an important, original, or

678 non-standard component of the core methods in this research? Note that if the LLM is used

679 only for writing, editing, or formatting purposes and does *not* impact the core methodology,

680 scientific rigor, or originality of the research, declaration is not required.

681 Answer: [Yes]

682 Justification: Section 2 and Appendix A identify LLM use for generation, certification, target

683 responses and scoring, including the retained GEPA seed and overlapping auditor/judge

684 roles.

685 Guidelines:

- 686 • The answer [N/A] means that the core method development in this research does not
- 687 involve LLMs as any important, original, or non-standard components.
- 688 • Please refer to our LLM policy in the NeurIPS handbook for what should or should not
- 689 be described.